

Summary of "The First Habitable-zone Planet From TESS"

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Abstract

TESS has discovered TOI 700 d, an Earth-sized planet in the habitable zone of its host star. It receives about 86% of the sunlight that Earth does. The planet orbits a small M-dwarf star about 100 light years away and is one of the first Earth-sized habitable-zone planets identified by TESS.

Introduction

The discovery of TOI 700 d provides valuable insights into planetary formation and the potential for life beyond our solar system. It shows TESS's effectiveness in detecting exoplanets in habitable zones.

Equation

The equilibrium temperature of a planet can be estimated using:

$$T_p = T_* \left(\frac{R_*}{2a} \right)^{1/2} (1 - A)^{1/4}$$

where T_p is the planet's temperature, T_* the star's temperature, R_* the radius of the star, a the orbital distance, and A the albedo.

Table

Table 1: Parameters of TOI 700 d

Parameter	Value
Radius	1.19 Earth radii
Orbital Period	37.4 days
Stellar Type	M dwarf
Insolation	0.86 $S_{\oplus}$
Distance from Earth	~100 light-years

Figure

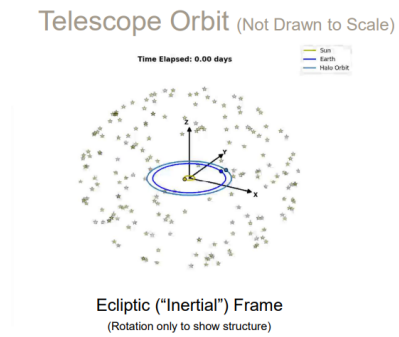


Figure 1: Illustration of the TOI 700 planetary system. (Image: NASA)